

Dhruv Agarwal

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Technical Skills

Languages and DataBases: Python , C/C++, Pandas, Numpy, SQL, MySQL

Visualization Tools:Power BI, Matplotlib / Seaborn

AI & Tools:Generative AI, Prompt Engineering, LangGraph, LangChain, Vapi AI, Google Gemini

Frameworks & Stack:Next.js, TypeScript, FastAPI, Firebase, Tailwind CSS, React Hook Form, Zod

Other Skills:Collaborative, Quick learner

Experience

- Fire AI**
AI Engineer Intern
 - Performed end-to-end QC verification of AI-powered financial dashboards for Venture Catalysts portfolio companies, comparing metrics against AWS S3 source data to detect scale errors, wrong mappings, and data mismatches across 10+ companies.
 - Built Python data validation pipelines using pandas and openpyxl to parse XLSX, XLS, and PDF files, automating metric comparison across monthly, quarterly, and annual cadences.
 - Audited AI agent conversation logs using a structured evaluation framework, assessing orchestrator routing, guardrail accuracy, and response quality; delivered JSON audit reports to the engineering team.
 - Analyzed chatbot conversation datasets to classify response quality (Exact, Partial, No Response) and produced humanized analysis reports with user intent tagging and feedback observations.
- AppSquadz Software Pvt.Ltd.**
Software Developer in Training
 - Developed interactive Power BI dashboards to visualize key performance metrics, cutting manual reporting time by 50%.
 - Automated data ingestion and cleansing pipelines using Python (pandas, NumPy) and AI techniques, ensuring consistent and reliable analytics data.
 - Implemented responsive UI components with HTML5 and CSS3, enhancing user engagement and accessibility across multiple devices.
 - Designed and optimized relational database schemas in MySQL; wrote SQL queries and stored procedures to support backend services.
- Eastern Software Solutions Pvt.Ltd.**
Machine Learning Intern
 - Worked on **NeuraTask**, an **AI-powered automation system** designed for intelligent workflow optimization in business processes.
 - Gained exposure to **Machine Learning techniques** and their application in workflow automation.
 - Assisted in developing modules under the guidance of the **Senior Project Manager**, ensuring efficiency in task assignment and process optimization.
 - Contributed to documentation, research, and testing of system features.

Feb 2026 – Present

Mumbai, India

July 2024 – May 2025

Noida, India

May 2025 – July 2025

Noida, India

Personal Projects

- Intvu — AI Interview Agent** | Next.js 15, TypeScript, Vapi AI, Google Gemini, Firebase, Tailwind CSS
Built a full-stack AI-powered mock interview platform featuring voice-based interviews conducted by a Vapi AI agent, with real-time live transcripts and a countdown timer during sessions.
Integrated Google Gemini via AI SDK to grade interview performance across 4 categories and generate structured feedback; added PDF resume upload to extract skills and personalize question sets.
Developed 8 features including a Practice Mode (text-based Q&A with per-answer AI feedback), Analytics Dashboard with score trend charts, a Top-20 Leaderboard, and one-click LinkedIn/Twitter share.
Implemented Firebase Auth with session cookies, Next.js middleware for route protection, and a downloadable HTML feedback report (print-to-PDF ready).
- SecureRAG — Conversational AI with Guardrails** | Python, OpenAI API, FAISS, Sentence Transformers, Streamlit
Built a production-grade two-agent RAG system with dual-layer security: a Guardrail Agent (Unicode-

normalised regex scan + few-shot GPT classifier with 10 labeled examples) blocks prompt injection and jailbreak attempts before queries reach the LLM.

Implemented FAISS vector store with Sentence Transformers embeddings and cosine similarity search; added indirect injection scanning on every retrieved chunk to defend against RAG-poisoning attacks.

Engineered a context-aware Main Agent with token-aware history trimming (8 turns / 1800 tokens), streaming output, and blocked-turn isolation so malicious queries never re-enter conversation context.

Applied advanced prompt engineering: few-shot prompting, instruction tuning with 5 behavioural layers, structured output formatting, and silent chain-of-thought reasoning in system prompts.

- **Wine Quality Predictor** | Python, TensorFlow, Streamlit, NumPy, Scikit-learn

Built an AI-powered web app to predict wine quality using chemical properties. Trained a neural network on a real-world dataset, achieving accurate classification through preprocessing, scaling, and quantile-based labeling.

Integrated interactive visualizations and real-time prediction via Streamlit UI, enabling user-friendly quality assessments without lab testing.

- **Travel AI Agent** | Python, FastAPI, LangGraph, OpenAI API

Developed an AI-powered travel assistant using FastAPI and LangGraph, integrating OpenAI's API for intelligent trip planning and conversational capabilities.

Enabled real-time itinerary generation and destination recommendations with token-based verification and secure API handling.

Awards & Achievements

- **Advance Python Programming — Udemy**

Completed comprehensive Python programming course covering fundamentals, data structures, and basic automation.

- **Cloud Computing** from Google Developer Student Club

- **Internship Certificate**

Received from **India Space Lab (ISL)** for successfully completing an internship program where I learned about the role of software in **aerospace and satellite operations**.

- **Internship Certificate**

Received from **AppSquadz** for successfully completing an internship program.

- **Eastern Software Solutions Pvt.Ltd. —**

Received from **Eastern Software Solutions Pvt.Ltd.** for successfully completing an internship program.

Education

- **Jaypee Institute of Information Technology Sep 2022 – May 2026(Expected)** *Bachelor of Technology in Computer Science Noida, Uttar Pradesh*
- **The Khaitan School 2008 – 2020** *10th Board CBSE Sec-40, Noida*
- **The Vivekanand School 2020 – 2022** *12th Board CBSE Narela, Delhi*